

Aiming for Excellence, Persisting in Reliable Solutions.

Company Profile

April 2026



Snapshot

Founded

2004.09

Chairman

T.C. Lin

President

Steven Wang

Employees (as of Mar. 2026)

450

Headquarter

**Hsinchu,
Taiwan**

Global Locations

HTSI has established a global presence across Taiwan, China, Japan, Malaysia, Singapore, and the United States, providing timely and professional technical support through localized services.



Global Locations

Tainan (TW)

Thin Film & Membrane Production / R&D Line



Hsinchu (TW)

Headquarter / CPC Fab

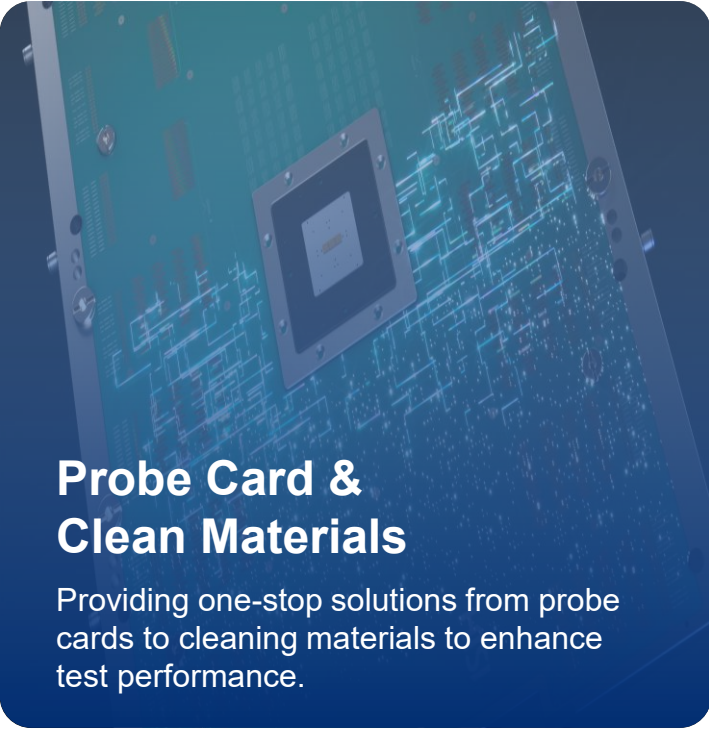
- Daily operations, logistics planning, and R&D
- Equipment Service Site
- Customized Products Development
- Probe Card Manufacture Line

Zhudong Fab

- SMT Manufacture Line
- Warehouse



Business Portfolio



Probe Card & Clean Materials

Providing one-stop solutions from probe cards to cleaning materials to enhance test performance.



Test Equipment & Engineer Service

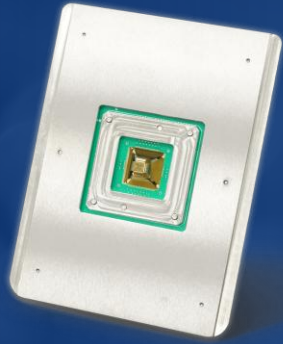
Full value-added services include sales(China), installation(Taiwan), and technical support



Semiconductor Equipment / Customized Products

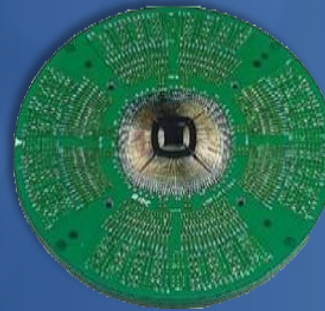
Develops and distributes semiconductor test equipment to enhance yield and provide customized solutions.

Probe Card Overview



Membrane Probe Card

- Supports 5G/6G, RF, and Filters applications
- 100% Taiwan-made for superior quality and delivery reliability
- Exclusive patented probe card leveling for stable wafer testing



Cantilever Probe Card

- Supports WAT, Logic, LCD Driver & GaN/SiC chip testing
- Technology transfer partnership with MJC, a top 3 global probe card manufacturer
- Custom designs for diverse product needs



Vertical Probe Card

- Cobra probe card: 100% Taiwan-made
- MEMS probe card: Partnership with MJC, integrated with HTSI components for real-time local support

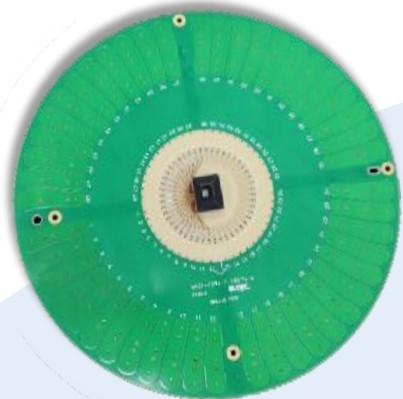
Cantilever Probe Card

Features

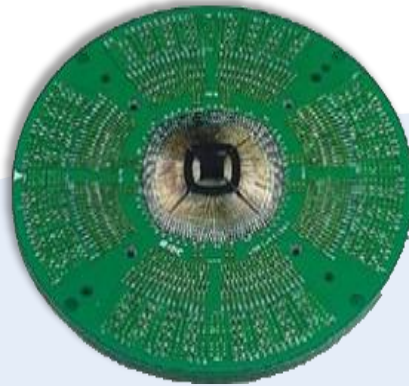
- 30A max current for high-power testing
- 1500V max voltage for high-voltage devices
- 30*30 μ m min pad size for fine-pitch probing

Device Categories

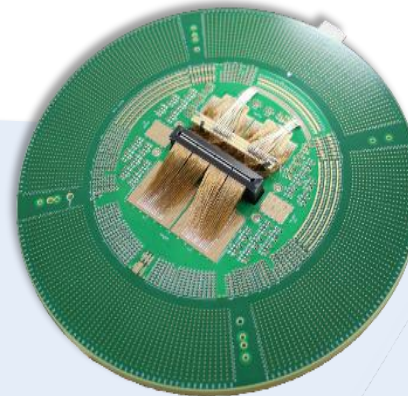
WAT



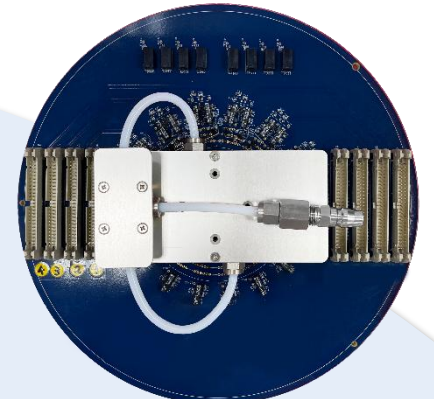
Logic



LCD-Driver



GaN/SiC



Cantilever Probe Card Capability

Max. pin count (pin)	3500
Max. current (A)	30
Max. voltage (V)	1500
Min. pad size (μ m)	30*30
Max. frequency (GHz)	2
Max. CCC/pin (A)	2
Min. pitch (μ m)	45

Vertical Probe Card

MEMS-SP (MJC)

Max. pin count: 50000 pins

Frequency: ~20GHz

	For PAD	For Solder Bump	For Cu Pillar Bump
Minimum pitch (μm)	80~130	130	85~130
Temperature ($^{\circ}\text{C}$)	-40 to 150	-40 to 150	-40 to 150
Tip Diameter (μm)	7 ± 3	72 ± 3	35~75
Contact force (g / mil)	0.7	0.9	0.5
Max OD (μm)	200	200	200
C.C.C (A)	1.2	1.6	1.2~2

MEMS-V (MJC)

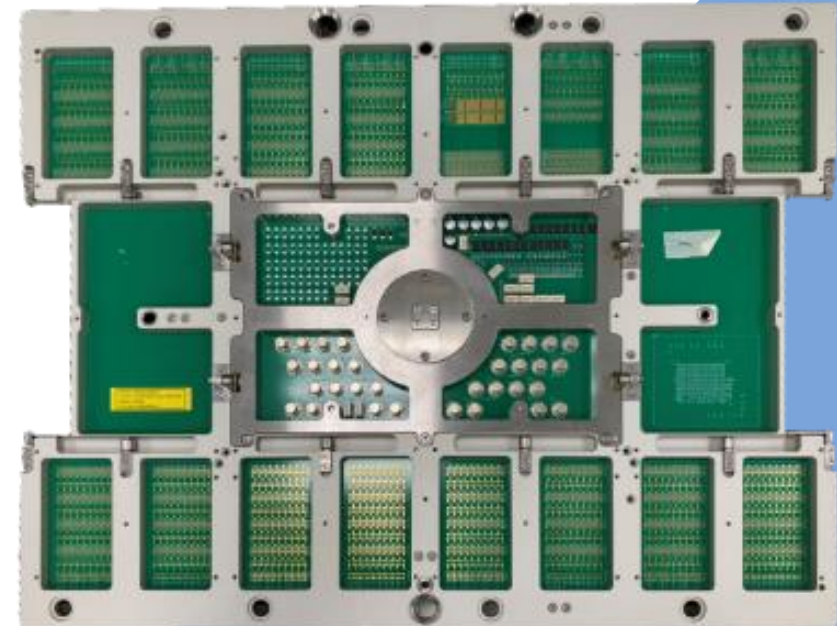
Max. pin count: 15000 pins

	MEMS-V Probe 50
Minimum pitch (μm)	50 μm (25um staggered)
Temperature ($^{\circ}\text{C}$)	-40 to 175
Tip Diameter (μm)	6 ± 2
Contact force (g / mil)	0.75
Max OD (μm)	100
C.C.C (mA)	400

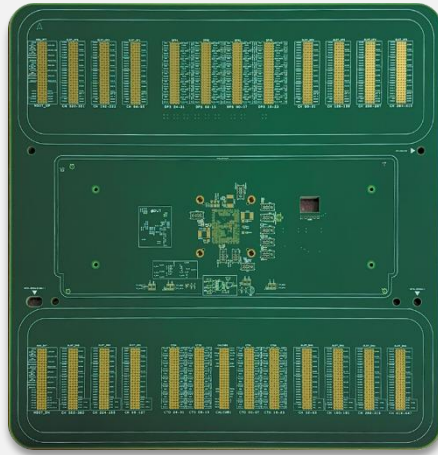
Cobra Probe Card

Max. pin count: 18000

Min. pitch: 80um

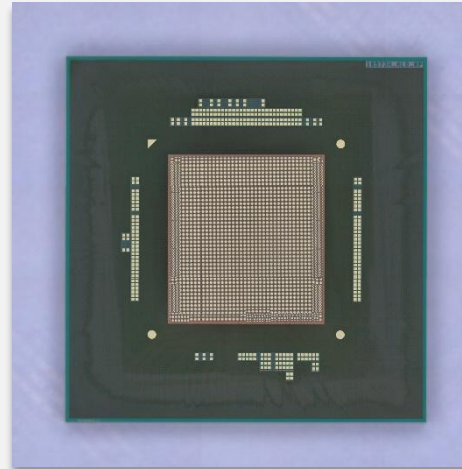


From Board to Probe



PCB

Design
Layout Capability



Substrate: MLO/MLC

MLO: 20/24/20
MLC: 30 layers
Line space: 12/12
Size: 100*100 mm



Probe Head

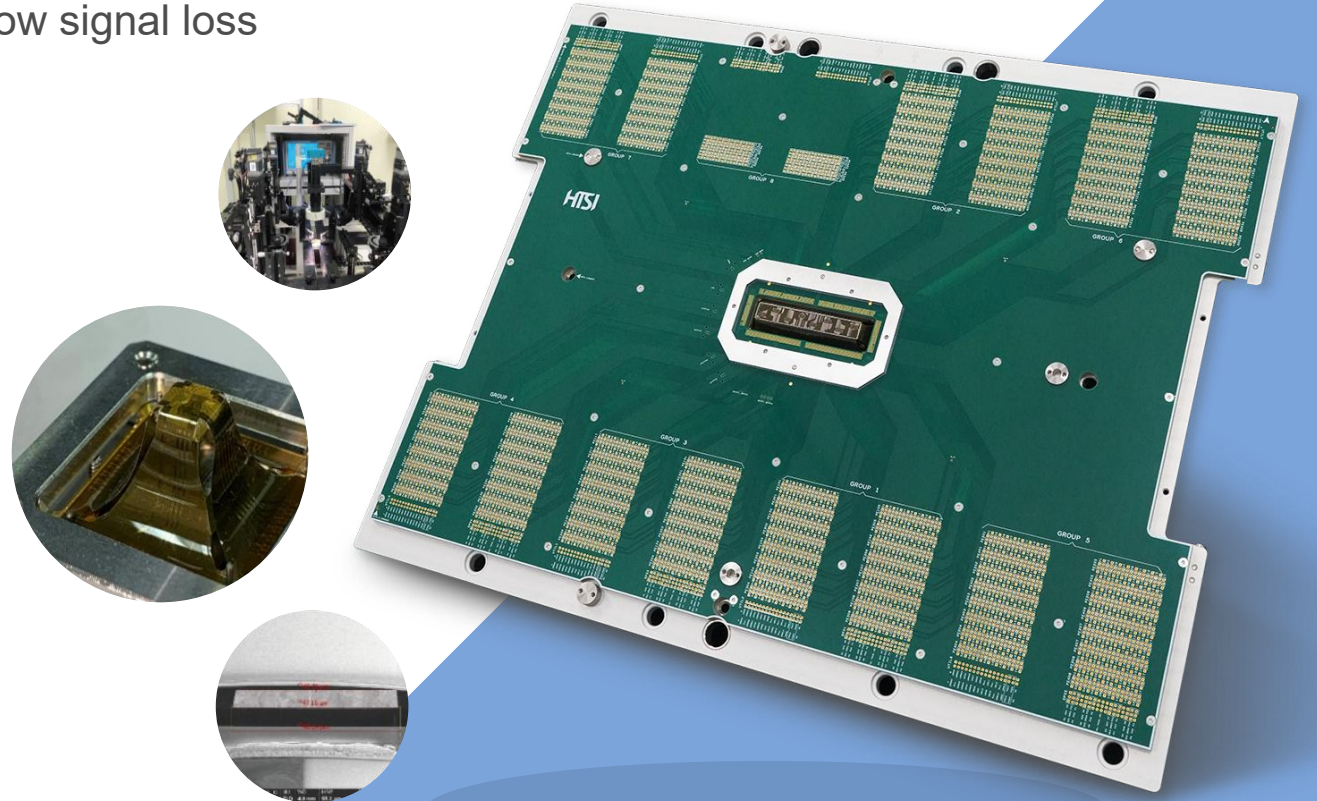
Fine pitch 50 um (Staggered 25 um)
Force down to 0.5 g/mil
Max. pin counts: 50,000

Membrane Probe Card

Features: GSG type measurement bandwidth can exceed 67GHz

- Probe mark uniformity and consistency
- Stable contact with well-controlled contact force
- 100% inhouse manufacturing and quick service solution
- High-bandwidth fidelity RF measurement with low signal loss

Capability		
Max. pin count (pin)		5000
S11 (dB)		<-20
S21 (dB)		>-3
Frequency (GHz)		~67
Min. pitch (μm)		65
Min. tip diameter (μm)		20
Lifetime (MTD)		1



Prober & Probe Card Sales/ Service

*as of Mar. 2026

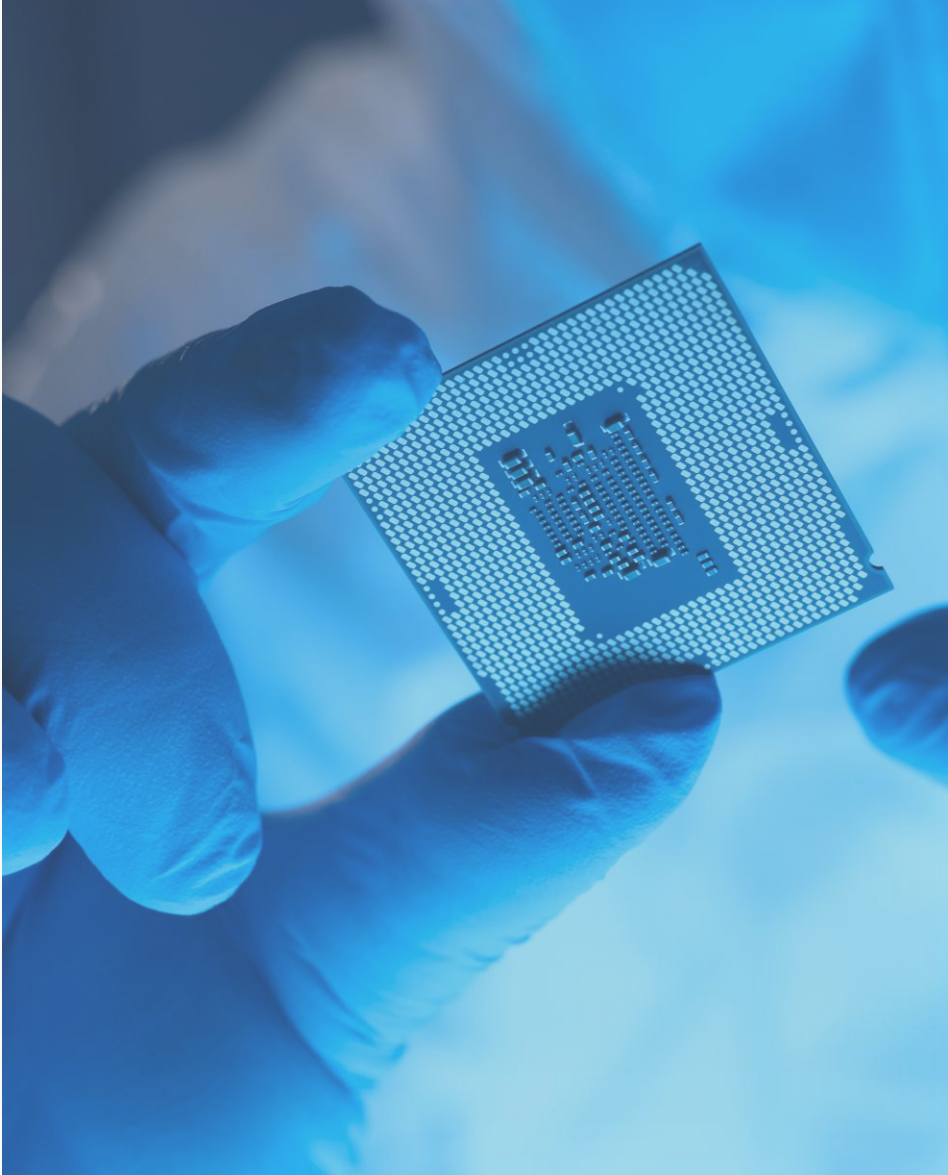
Area	Prober Field Eng.	NPI Eng.	Probe Card FAE
North + MI	75	12	12
Central	31	0	0
South	29	0	0
Total	135	12	12

Prober

- Sales/ Leasing Biz
- Prober Service & Support, CIP, etc.
- Quarterly/ Annual PM

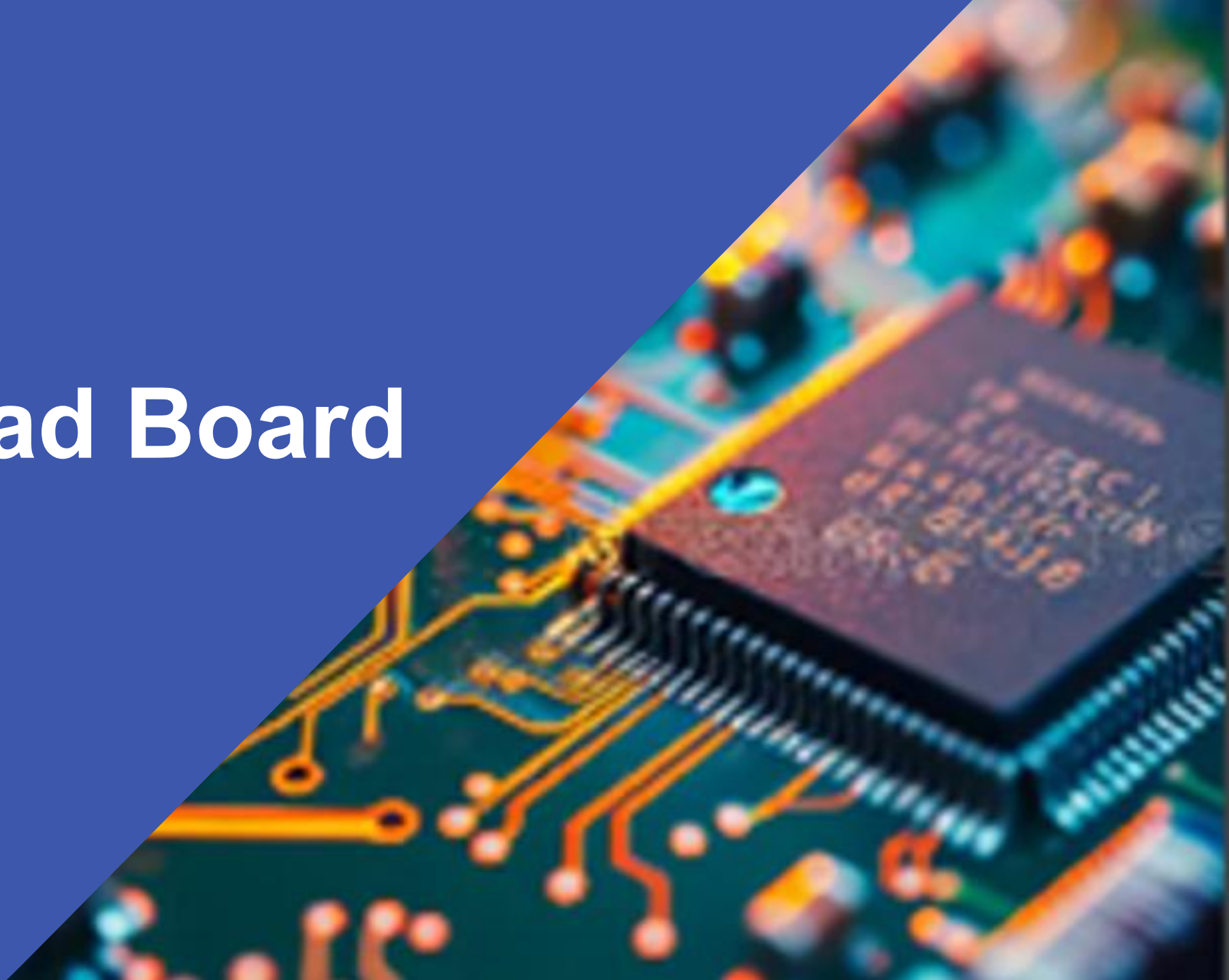
Probe Card

- Sales
- PM/ Rebuild/ FAE for CPC & VPC



HTSI Load Board

Overview



Design Resource

- Load board design location: Hsinchu
- Numbers of Load board design designers (simulation...etc.) and design capacity

Function	2026 H1(now)	2026 H2	2027
PCB/ST Design	10+3	16+3	35+3
SI / PI	2	3	3
FAE	3	3	4
Total	18	25	45

- Design tools used: Allegro (Layout) ; OrCAD Capture(Schematic) ; Ansys (SI / PI)

PCB design SPEC

Item	SPEC
最小間距 (Min Pitch)	0.28 mm
完成板尺寸 (Finished Board Size)	560*710 mm
層數 (Layer Count)	106 layer
板厚 (Board Thickness)	1~8 mm
板曲板翹 (Warp and Twist)	2 mil/inch
平坦度 (Flatness)	25 um/per inch
最小鐳射鑽孔尺寸 (Minimum Laser Drill size)	4 mil
最大縱橫比(鐳射孔) (Maximum Aspect Ratio(laser drill))	0.8
金厚 (Gold Thickness((um))	3~ 10 u"
最小線寬/線距(mil)(Minimum line width/line spacing)	2.5/2.5 mil
最大背鑽殘桿mil (Stub)	12 mil
最大縱橫比(通孔) Maximum Aspect Ratio(thru)	45:1

*Discussed by case

— About Assembly Ability —

Item	Min	Max
PCB Size	70mm * 50mm	710mm * 580mm
Thickness	1mm	8mm (13mm)
Component	01005	200mm * 125mm * 25mm
Ball mount	0.2mm	0.76mm
Repair	NA	70mm * 70mm

- 尺寸小於最小規格產品可使用載具方式生產

ATE Platform Experiences

Platform						
Teradyne	Advantest	CHROMA	YTEC	YOGOKAWA	Acco	Sigurd
Ultraflex	V93K Exa	C3650	V50	ST6730	STS8200	MAP
Ultraflex Plus	V93K_DD	C3680	S100_7D_by 8			
J750EX	T5375	C3380				
J750HD	T5377					
	ND3					
	ND4					

PCB Material

Material		
EM827	N4000-13	Megtron_6
Megtron_7	RO4350B	Meteorwave 2000
Meteorwave 4000	TU883	TU865
TU933	FR4_HTG	FR5

Simulation Experiences

Simulation	Item	
Simulation item	• TDR • S-parameter • Eye Diagram	• PDN • IRDROP
Simulation tool	• ANSYS Siwave • ANSYS HFSS • ANSYS Electronics Desktop	
Simulation interface	• USB3 • HDMI • DDR4 • TYPE-C	

● Required Documents for PCB Simulation:

*  .sp *  .inc

SMT Process (Applicable for ODM/OEM)



IQC-進料檢驗



烤箱烘烤



Printer-My700



Mounter-Sipace X 4S



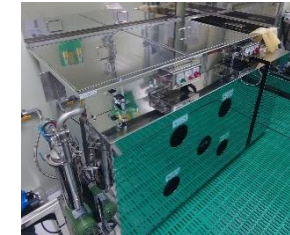
Reflow-HF326



Inspection



DIP一次焊接、組裝



超音波清洗製程



DIP二次焊接、組裝

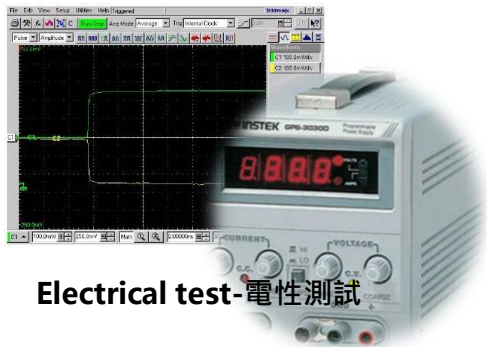


In
Process
Quality
Control

IPQC-製程品質管控



ICT-SPEA 4080L

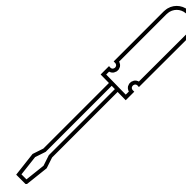


Electrical test-電性測試



Final
Quality
Control

FQC-成品品質管控

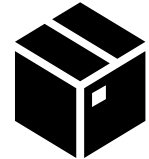


Stiffener、配件組裝



Outgoing
Quality
Control

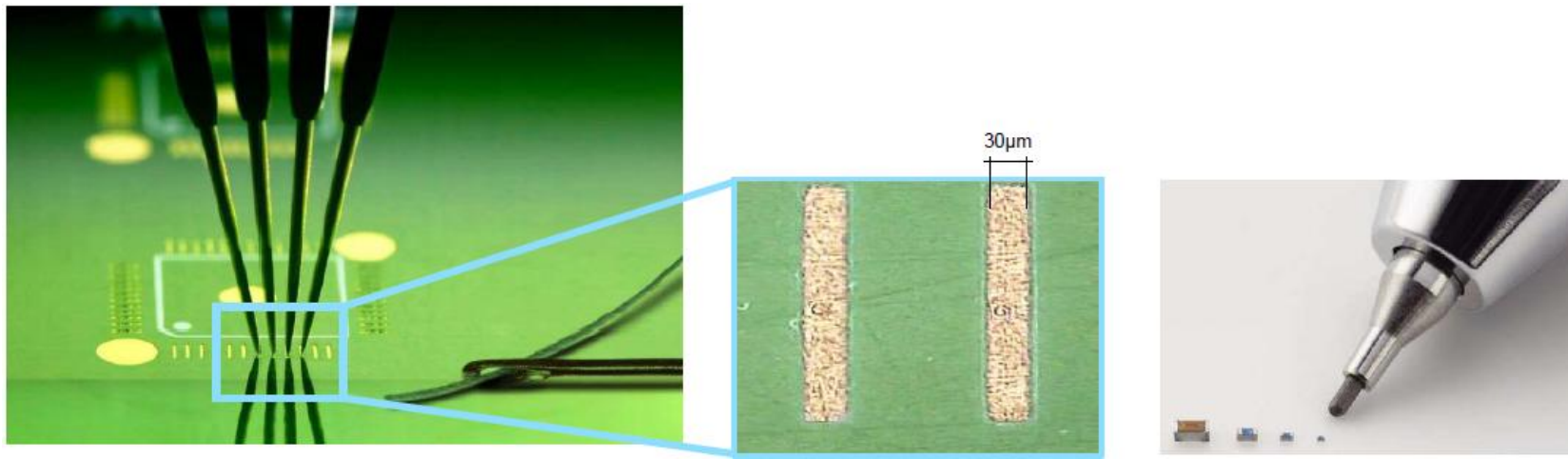
OQC-成品品質管控



包裝入庫

4080L Large Board Flying Probe Tester

TechSpecs	
Max measurable voltage	± 100 V
Max Contactable area	710x662mm
Min. Pad size	30 μ m (in ultra-accurate probing configuration)
Min. Pitch	160 μ m
Resonance Frequency	>50
Max Warpage@ fullspeed[μ m]	<10
Damping Ratio [%]	0.5



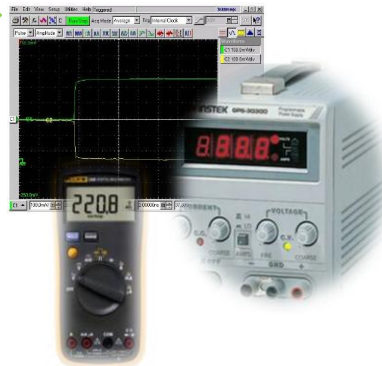
Repair Process



IQC-進料檢驗



導通絕緣測試



Electrical test-電性測試

分析問題、判斷解決方式



烤箱烘烤



BGA Rework Station



鐵板燒

選擇解決工具與方式



高週波烙鐵



導通絕緣測試



Electrical test-電性測試

重工後復測異常與判斷是否全檢

Outgoing
Quality
Control

OQC-成品品質管控



包裝入庫

How We Support Your Success



Collaborative Engineering Support

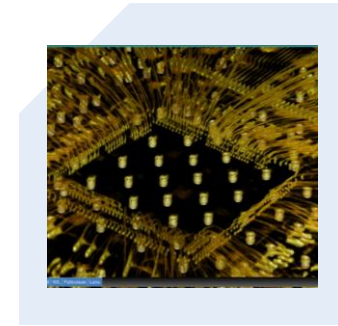
Design, simulation, and manufacturing of custom ATE. (VPC/CPC/LB/BIB/EVB).



Engineering & Assembly Services

Gerber review, stack-up consulting, and layout optimization.

High-precision SMT assembly for prototyping and production.



Non-Destructive IC Analysis

Advanced 3D X-Ray inspection for internal IC structure analysis and defect detection.

HTSI

漢民測試
Hermes Testing

THANKS

HTSI